Adversarial Robustness in Network Intrusion Detection Systems

via Attribution-Guided Feature Selection and Dynamic Multi-Stage Adversarial Training

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Presentation Outline

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Problem Statement

Modern Deep NIDS achieve **>95% accuracy** on normal traffic, but collapse to ~30% under tiny perturbation **[5, 9]**.

System Performance

Clean vs. Robust

95% → **~30%**

CLEAN ACC — ROBUST ACC

The Adversarial Evasion Mechanism

Attackers slightly alter flow timing and packet sizes.

Consequence: The AI is tricked into classifying dangerous attacks as normal benign traffic.

Gradient-Volatile Features

Fragile Features: Classifiers rely on weak traffic clues that attackers can easily disguise.**[5]**

Transient Socket Shortcuts

Shortcut Learning: Models memorize port numbers and IP addresses instead of learning real attack behavior.**[18]**

Defensive Hardening Trade-Offs

Stale Defenses: Existing defenses train once on old attacks, making them helpless against adaptive, modern attackers.**[1]**

Motivation & Objectives

Building fast, accurate, and honest AI defense for live enterprise networks

01

Real-Time Speed

High-speed networks cannot wait: slow detection drops valid packets.

Goal: Overcome multi-millisecond defense delays (**85–220 ms** in **[3]**, **>2.4 ms** in **[4]**) that violate line-rate packet forwarding."

02

Accuracy vs Robustness

Defending models usually causes a 2% to 6% drop in normal accuracy.**[2, 17]**

Goal: Stop attacks without hurting normal accuracy by removing noisy features.

03

Benchmark Hygiene

Goal: Completely remove all port and IP shortcuts so the AI learns real attack behavior.**[18]**

Enforce strict partitioning and class balancing.**[15]**

Literature Review

Published defense paradigms, benchmark methods, and operational trade-offs.

REF	PUBLISHED METHOD	DEFENSE PARADIGM	BENCHMARK RESULT	CRITICAL BOTTLENECK
[1]	**Static Triad Augmentation**	Offline Static Augmentation	`88.0% Defended`	**Offline attack staleness;** retained port shortcuts
[2]	**AMDS (Adaptive Multi-Stage)**	Dynamic Minimax PGD Hardening	`89.4% PGD$_{20}$`	**2.1% clean drop;** unpruned 78-dim overhead (+140% GPU)
[3]	**XAI-Centroid Gate**	Runtime Attribution (SHAP)	`88.5% Detection`	**Severe latency (85–220 ms);** violates line-rate
[4]	**CSAGC-IDS**	Channel-Spatial Graph Attention	`76.2% Robust`	**Buffering delay (>2.4 ms);** graph topology spoofing
[5]	**Randomized Smoothing**	Certified Provable L_2 Radius	Provable Limit	**Clean accuracy crash:** 94.5% → 76.2% (-18.3%)

Key Literature Finding Existing defenses force an unresolved trade-off: offline methods suffer attack staleness, runtime defenses violate line-rate throughput, and certified methods severely penalize clean traffic.

Literature Review

Three structural vulnerabilities identified across existing NIDS defenses.

01

Perturbation Staleness

`Offline Augment. [1]`

MECHANISM

Defenses like Static Triad inject pre-computed static attack samples once before retraining.

VULNERABILITY

As neural weights evolve during training, the decision boundary shifts away from static samples, allowing dynamic adaptive attacks to easily evade the model **(16%–20% evasion rate)**.

02

Line-Rate Latency Barrier

`Runtime Defense [3, 4]`

MECHANISM

Complex runtime verifications like real-time SHAP gates (85–220 ms) [3] and graph batch buffering (>2.4 ms) [4] introduce heavy computational overhead.

VULNERABILITY

High-speed cloud switches operate on microsecond scales **(<0.05 ms)**; multi-millisecond per-flow processing causes buffer overflow and packet drops.

03

Shortcut Learning Bias

`Feature Selection [1, 18]`

MECHANISM

Conventional feature selection evaluates feature importance exclusively on benign, nominal traffic.

VULNERABILITY

Models retain ephemeral socket shortcuts **(Src/Dst Port, IP)** and gradient-volatile features that attackers trivially manipulate to bypass detection.

Comparative Analysis

State-of-the-art benchmark frontiers: Every published defense faces a critical trade-off.

BENCHMARK FRONTIER	METHOD & REF	BEST RESULT	DEFENSE MECHANISM	LIMITATION
Best Clean Accuracy	**Deep MLP [7]**	**98.2%** Clean	Standard Deep MLP / Graph Attention	**Zero Defense:** Collapses to **<32%** under white-box PGD attacks; or incurs 2.4 ms graph delay.
Best White-Box Robustness	**AMDS [2]**	**89.4%** PGD_{20}	Multi-Stage Dynamic PGD	**Clean Drop & Overhead:** Suffers **2.1% penalty**; keeps 78 dimensions (+140% GPU cost).
Best Static Defense	**Static Triad [1]**	**88.0%** Defended	Static Triad Augmentation	**Staleness & Shortcuts:** Offline attacks go stale (**16–20% evasion**); retained port shortcuts.
Best Black-Box Robustness	**Query Anomaly [6]**	**88.6%** Black-Box	Query Perturbation & Anomaly Tracking	**High Complexity:** Stateful query tracking required; vulnerable to adaptive white-box transfer.
Best Certified Defense	**Rand. Smoothing [5]**	**Provable L_2** Radius	Randomized Smoothing	**Severe Accuracy Crash:** Clean accuracy drops from **94.5% → 76.2%** (-18.3% penalty).

The Literature Dilemma

Across published literature, no existing system achieves high robustness without suffering a severe penalty in clean accuracy, latency, or computational complexity.

Proposed Methodology

Data Sanitization and Screening

01 **Cleaning** — Feature Drop

02 **Partitioning** — Train/Val/Test

03 **MI Screening** — Shannon MI

04 **SMOTE** — Balancing

Data Cleaning

Permanently remove non-generalizable features: **Src/Dst IP, Ports, Timestamp, Flow ID**.

Result: 69 continuous features remain.

Data Partitioning

80% Dev Set (5M flows) / **20% Test Set** (1.2M flows) locked Day 0.

Dev split: **72% Train / 8% Val**.

Parameters fitted strictly on 72% Train.

Feature Selection

Mutual Information filters 69 down to **top 40 candidates**.

Accelerates Shapley computation by **>60%**.

Oversampling

Continuous-Only SMOTE**[15]** applied **strictly within training split** (72% Train).

Balances minority attacks without interpolating discrete flags.

Proposed Methodology: SHAP

Our Approach: Measure feature stability under attack to keep only the most reliable features using SHAP. [11]

01 Clean-Only Vulnerability

Why Normal Feature Selection Fails:

Features that look important on normal traffic are often the easiest for attackers to manipulate.

02 Dual-State Attribution Auditing

- Train baseline canary MLP on clean data using 40 candidates.
- Attack a baseline model with PGD[8], then compare feature importance on normal vs. attacked traffic.

03 Exact Mathematical Stability Score (S_j)

Quantifying individual feature robustness under perturbation:

$$S_j = 1 - \frac{\left|\bar{\phi}_j^{\text{clean}} - \bar{\phi}_j^{\text{adv}}\right|}{\max\left(\bar{\phi}_j^{\text{clean}} + \bar{\phi}_j^{\text{adv}},\ 10^{-10}\right)}$$

04 80% Cumulative Stability-Mass Selection

Rank features descending and select minimal subset **m** satisfying:

$$\min m \quad \text{s. t.} \quad \sum_{j=1}^{m} S_j \geq 0.80 \cdot S_{\text{total}}$$

27 STABLE FEATURES SELECTED

13 VOLATILE FEATURES PRUNED

Proposed Methodology : Two-Phase Dynamic Training

Curriculum training to mitigate static attack staleness.

Curriculum Optimization Rationale:

Why Two Phases? Attacking a brand-new model from day one ruins its learning. It must learn normal traffic patterns first before facing attacks.

PHASE 1

Representation Warmup (Epochs 1-10)

- **Phase 1 (Epochs 1–10):** Train on normal traffic first so the model builds solid, accurate foundations.
- Train exclusively on clean data using standard cross-entropy loss.

PHASE 2

Dynamic Online Training (Epochs 11-50)

- **Phase 2 (Epochs 11–50):** Generate live, fresh attacks for every training batch to defeat evolving attacks without hurting normal accuracy.
- Updates model parameters via a balanced joint objective.

Joint Objective: $L_{\text{total}} = 0.5 \cdot L_{\text{clean}}(f_\theta(x), y) + 0.5 \cdot L_{\text{adv}}(f_\theta(x_{\text{adv}}), y)$

Result: Mitigates static attack staleness while preserving benign representation anchors.

Dataset & Environment

Experimental configuration and technical architecture parameters

1. Benchmark Dataset

Standardized `dhoogla/csecicids2018` Parquet curation **[19]** (replaces raw csv to resolve corrupted headers and NaN flaws while guaranteeing 100% reproducibility).

Total Valid Flow Volume:

6,320,003

flows (post-sanitization).

Partition Breakdown:

72% Core Train: 4.55M flows
8% Validation: 505K flows
20% Frozen Test: 1.26M flows

2. Platform & Stack

Platform: AMD Radeon RX 7600 (8 GB VRAM, RDNA 3 architecture).

Software Stack: Python 3.12, PyTorch, ROCm Windows, Adversarial Robustness Toolbox.

Batch Optimization: Vectorized GPU tensor operations **(Batch Size = 2048)**, mixed-precision.

3. Target Architecture

Topology: 4-Layer Multi-Layer Perceptron (MLP).

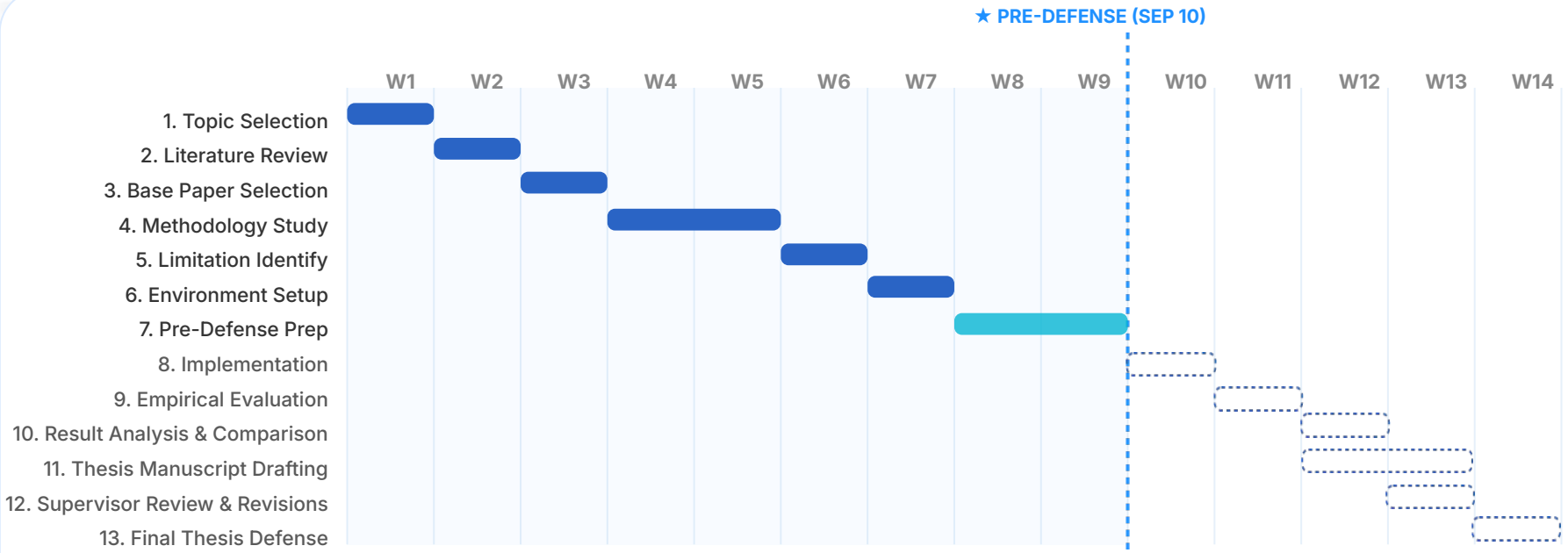
Layer Design: Linear → Batch Normalization → ReLU Activation → Dropout (p = 0.30).

Complexity: Compact feed-forward pass (~14,000 parameters) ensuring high GPU throughput.

27 → 128 → 64 → 32 → 2

Weekly Research Timeline & Gantt Chart

Project Execution Timeline (Weeks 1-14)



Current Status (September 10, 2026)

Tasks 1 through 7 are complete (Theoretical formulation, literature audit, data ingestion pipeline verified). The **Pre-Defense Milestone** has been successfully achieved.

Immediate Transitions & Next Steps

Immediately transitioning to **multi-threat evaluation** (Weeks 11–12) and **final manuscript submission** (Weeks 13–14).

Expected Results & Research Hypotheses

Comprehensive multi-threat evaluation protocol and performance targets.

Clean Accuracy

≥ 95%

Pruning 13 unstable features mitigates degradation.

Robust Accuracy

88.0% – 90%

PGD-100 (ε=0.05) vs. undefended baselines.

Attack Success Rate

< 12.0%

Under iterative white-box gradient attacks.

Inference Speed

< 0.05ms

Amortized per flow (preliminary: <0.02 ms).

Multi-Threat Evaluation Suite

First-Order Gradient Attacks
FGSM, PGD20, PGD100, BIM, DeepFool (minimum-distance boundary perturbation)

Parameter-Free & Optimization Attacks
Auto-PGD (APGD-CE), Carlini-Wagner (l∞)

Decision-Based & Black-Box Attacks
Square Attack (query-based), Surrogate Model Transferability

Gradient Obfuscation Safeguards
Validate that robustness is genuine and not caused by gradient masking (verifying non-zero input gradients and evaluating black-box transferability **[12]**).

Conclusion & Immediate Next Steps

Summary of proposed contributions and immediate project execution roadmap

PROPOSED CONTRIBUTIONS

Robust Feature Selection & SMOTE Balancing

Formulates Adversarial Stability SHAP (80% stability rule isolating 27 invariant features) combined with leak-free SMOTE class balancing strictly inside the training split.

Two-Phase Dynamic Adversarial Training

Proposes representation warmup (Epochs 1–10) followed by live 5-step online PGD mini-batch hardening (Epochs 11–50) to track moving decision boundaries and eliminate attack staleness.

Comprehensive Multi-Threat Evaluation

Rigorous robustness verification across diverse attack modalities (FGSM, DeepFool, PGD, CW, Square Attack) with zero socket-shortcut leakage.

SEPT 11 – OCT 11

Immediate Next Milestones

01 Execute 3-seed experimental runs across all 9 adversarial threat models.

02 Conduct ablation studies: (1) Full 69D vs. 40D MI vs. 27D Stable; (2) Static Triad vs. Dynamic PGD.

03 Complete and submit the final undergraduate B.Sc. thesis manuscript for defense.

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